Ecological Processes

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Species richness is not a good predictor for above-ground biomass in a warm temperate deciduous broadleaf forest

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Abstract

Background Biomass is the result of long-term production and metabolism in forest ecosystems and is an important indicator of the carbon storage capacity of forests. Although there is increasing empirical evidence supporting the positive impact of biodiversity on forest productivity and biomass, there is still uncertainty about the relative importance of tree diversity in determining carbon storage compared to other factors such as environmental conditions, functional characteristics and stand structure, especially in complex forest ecosystems.

Methods In this study, based on dataset from a 25-ha forest dynamics monitoring plot, we investigated the effects of tree diversity, environmental variables, functional traits and stand structural attributes on above-ground biomass (AGB). Spearman correlation coefficients were used to analyze the correlations between AGB and the variables. The relative importance of these factors in influencing AGB variation was assessed using a random forest model. Structural equation model was used to determine the direct or indirect effects of each factor on AGB.

Results The results showed a negative, though not significant, correlation between species richness and AGB. There was a significant positive correlation between leaf dry matter content and leaf tissue density, implying that more leaf photosynthetic products were utilized for dry matter accumulation. The variation in AGB was mainly explained by the maximum diameter at breast height and the coefficients of variation of diameter at breast height, suggesting that large diameter individuals contribute disproportionately to AGB. In addition, AGB was also influenced by topographic factors (i.e., altitude and slope), while there was no significant correlation with soil variables.

Conclusions This study reflects the response of AGB to different influencing factors. Our study emphasizes that stand structure attributes may be more suitable as predictors of forest AGB than species richness.

Keywords Above-ground biomass, Stand structure, Functional trait, Biodiversity, Selection effect, Qinling Huangguan

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Background

Forests are the largest carbon (C) pool in terrestrial ecosystems and play a key role in regulating the global C cycle. Forests absorb approximately 35% of the CO_2 emitted by fossil fuels, account for nearly 25% of the global carbon sink, and exhibit the highest biological productivity among terrestrial ecosystems (King et al. 2012). A large portion of this C is stored in above-ground biomass (AGB) and soil organic matter (Fahey et al. 2010). AGB, being susceptible to natural or anthropogenic influences (Masha et al. 2017; Zhai et al. 2022), constitutes a primary factor driving changes in carbon storage within forest ecosystems. Therefore, research on biomass storage and its influencing factors is critically important. In the past decades, the contribution of species diversity to forest biomass has received widespread attention for its close relationship with C sequestration (Yuan et al. 2019; Augusto and Boča 2022), but other factors were neglected.

Many studies on species richness, the most intuitively descriptive variable of biodiversity, reported a positive correlation with forest productivity (Cavanaugh et al. 2014; Liang et al. 2016; Chen et al. 2018; Huang et al. 2018; Augusto and Boča 2022). This positive effect is usually explained by the niche complementarity effects and selection effects. The niche complementarity effect refers to ecological differences among species leading to more efficient resource utilization within communities. Increased species diversity enhances the potential for positive interactions among species, thereby contributing to enhanced productivity (Tilman et al. 1997). The selection effect suggests that the higher species richness, the greater the chance that highly productive species will occur, and thus community productivity will increase (Huston 1997). Another mechanism consistent with selection effects is the mass ratio hypothesis, which supposes that ecosystem function is largely determined by biomass-dominant species (Grime 2002; Ruiz-Benito et al. 2014). For example, Garnier et al. (2004) argue that using only the functional traits of dominant species can reflect the main characteristics of an ecosystem. Since higher productivity means faster biomass accumulation, these mechanisms of community productivity also apply to biomass (Yue et al. 2020). Although many studies have reported that species richness can significantly increase forest productivity (Liang et al. 2016; Weisser et al. 2017; Chen et al. 2018; Huang et al. 2018; Jactel et al. 2018; Liu et al. 2018; Li et al. 2024), some studies reported that they are not correlated (Seidel et al. 2013) or even negatively correlated (Vilà et al. 2003; Forrester and Bauhus 2016). Plant functional traits play an important role in ecosystem function, which is the result of species' adaptation to different environments in the long-term evolutionary process, and can objectively express the adaptability of plants to external environment. (Vargas-Larreta et al. 2021). Some studies suggest that functional diversity and functional traits may be better predictors of forest biomass or productivity than species richness (Gross et al. 2017; Vargas-Larreta et al. 2021; Li et al. 2022). For example, Li et al. (2022) found that functional diversity was the dominant factor for ecosystem multifunctionality in plantation forests by comparing the results in monoculture and mixed forests (Li et al. 2022).

Higher species diversity promotes the coexistence of trees with different growth forms and resource-use strategies, which contributes to greater resource-use efficiency. Therefore, niche complementarity effects are inferred when there is higher biomass and productivity in communities with high diversity (Mensah et al. 2018). On the other hand, due to the asymmetric competitiveness of tree sizes, taller trees may affect the growth rate of smaller individuals in their neighbors (Chi et al. 2015), and the selection effects occur when the specific traits of dominant tree species within a community have a greater contribution to ecosystem processes (Mensah et al. 2018). The coexistence and growth of trees of various species and sizes within distinct spatial ecological niches rely on physiological traits associated with life form and shade tolerance, as well as their competitive ability to acquire resources under environmental pressures (Jucker et al. 2015; Ehbrecht et al. 2021). Therefore, both hypotheses may simultaneously determine the impact of diversity on ecosystem functioning (Carroll et al. 2011; Lasky et al. 2014). Large trees contribute significantly to aboveground biomass, so even with similar species richness, biomass can vary depending on the proportion of large trees. When the proportion of large trees is substantial and smaller individuals primarily contribute to diversity, this may obscure the positive relationship between diversity and biomass. These hypotheses about diversity and ecosystem functioning have been tested repeatedly, although there is consensus that ecosystem functioning (e.g., biomass, productivity, and carbon storage) tends to increase with biodiversity. However, it is not known whether this positive correlation is enhanced or weakened by environmental differences in different regions. In addition, the relative importance of two mechanisms, ecological niche complementarity effects and selection effects, may vary depending on the choice of scale (Gonzalez et al. 2020), local environmental conditions, and forest type.

Notably, environmental variables are also major moderators of AGB, because they determine to some extent the availability of resources for plant growth and survival (Masha et al. 2017; Yuan et al. 2018). The local environmental filtering hypothesis states that ecological

distinctiveness is related to local environmental conditions (e.g., topography, soil nutrients) that determine subtle changes in forest structure and composition (Costa et al. 2005; Yao et al. 2021). Soil properties may influence the creation of competitive advantage or niches, thereby altering the composition and abundance in the community (Paoli et al. 2006). Furthermore, topography and soil conditions can indirectly affect forest function by influencing species diversity and structural diversity (Condit et al. 2013; Poorter et al. 2017; Ali et al. 2019). Thus, when investigating the factors that influence AGB, these variables must be taken into account.

Although these factors mentioned above all have effects on AGB, their relative importance remains unclear. In this study, we used the census data in 2019 from a 25-ha forest dynamics monitoring plot in Huangguan of the Qinling Mountains, to determine the relationship between AGB and species diversity, plant functional traits, stand structure, and environment. To investigate the effects of microenvironments, such as terrain and soil, at a fine scale, we chose to conduct our analysis at a 20 m × 20 m scale. We ask the following three major questions: (a) Does species richness have a significant positive effect on AGB and support the niche complementarity effects? (b) Do functional traits better explain differences in AGB compared to species richness? (c) What are the key factors that contribute to the variations in AGB on the 20 m × 20 m scale?

Materials and methods

Study site

The study area is conducted in a 25-ha (500 m × 500 m) forest dynamics monitoring plot (referred to as Qinling Huangguan plot; 33.54°N, 108.37°E) in the Huangguan Management Station of Changqing National Nature Reserve, Shaanxi Province, China. The reserve is located in the south slope of the middle part of the Qinling Mountains, with a total area of 12,372 ha, and an altitude of 880–2340 m. The Qinling Huangguan plot is a typical warm-temperate deciduous broad-leaved forest. The vegetation belongs to the transition zone from evergreen broad-leaved forest in the northern subtropical to deciduous broad-leaved forest in the southern warm-temperate zone. The region has an annual average precipitation of 908 mm, with rainfall predominantly occurring from July to September. The annual average temperature is 12.3 °C (with a recorded maximum of 36.2 °C and minimum of −13.1 °C). The average temperature in the hottest month (July) is 23.3 °C, and in the coldest month (January) is 0.6 °C. The average frost-free period lasts for 216 days per year (He et al. 2022). The altitude of the plot ranges from 1280.3 m to 1581.8 m above sea level, with an average altitude of 1414.2 m and a maximum elevation difference of 301.5 m.

Data collection

The Qinling Huangguan plot was constructed in May 2019 and the first community survey was completed in September 2019. According to the technical specifications of Chinese Forest Biodiversity Monitoring Network (CForBio) and Center for Tropical Forest Science of Smithsonian Tropical Research Institute (CTFS), the 25-ha plot was divided into 625 plots of 20 m × 20 m using a total station, and each 20 m × 20 m plot was further divided into 16 small 5 m × 5 m plots. For each 20 m × 20 m plot, topographic variables (altitude, slope, convexity) were calculated. We determined species name, spatial coordinates, and measured diameter at breast height (DBH) for each individual stem with DBH ≥ 1 cm. There were 75,137 individual woody plants (95,679 including branches) with DBH ≥ 1 cm, belonging to 44 families, 83 genera and 121 species.

Above-ground biomass calculations

We selected all individuals with a DBH ≥ 4 cm for the calculation of AGB. Tree biomass for each individual was estimated from stock volume using the biomass expansion factor method (Fang et al. 2001, 2014). First, We calculated the volume of timber for different types of tree species using the one-dimensional standing timber volume equation: $V = V_0 + (D - D_0) \times \Delta V$, where V is the volume of standing timber, D is the DBH of the tree, V_0 is the volume of timber corresponding to the integer portion of DBH, D_0 is an integer rounded down to the nearest integer of DBH, and ΔV is the incremental volume of timber when the DBH is increased by 1 cm. Then, we calculated the biomass per unit area of each tree species within each 20 m × 20 m plot according to the biomass expansion factor function: $y = ax + b$, where y is the biomass per unit area (t/ha), x is the volume per unit area (m^3/ha), a and b are parameters (the values of different tree species are shown in the Table S1). Finally, we calculated the total biomass of all trees within each 20 m × 20 m plot. For more detailed information on AGB calculations, see the Appendix S1.

Environmental variables

The environmental variables included topographical and soil factors. We superimposed a regular grid of 20 m × 20 m quadrats on the study area and calculated four topographic variables (altitude, slope, convexity, and aspect) for each plot. The slope direction is treated sinusoidal. With reference to the CTFS standard for soil sampling, 972 soil sampling points were taken for soil nutrient analysis, excluding points covered by boulders or with

tree roots that were difficult to be sampled. A total of four basic soil indicators were obtained: soil organic matter content (SOM), pH, soil total nitrogen content (TN), and soil total phosphorus content (TP). We used Kriging spatial interpolation with ArcGIS software to obtain the mean value of each soil chemical index within each 20 m × 20 m plot.

Indicators of biodiversity and functional traits

In July–August 2022, leaves were collected from woody plants in the 20 m × 20 m plots. Three to five mature, well-established adult plants per species were selected for sampling, and a total of 556 leaf samples were collected from 119 species. Mature leaves exposed to sunlight were chosen for measurement whenever possible. Leaf fresh weight (LFW) and leaf thickness (LT) were measured immediately on the day of leaf collection. LFW was determined by weighing each numbered labeled leaf using an electronic balance (accuracy 0.0001 g). The leaves were scanned using Epson scanner and leaf area (LA) was calculated using ImageJ software. LT was measured using an electronic digital vernier caliper with an accuracy of 0.01 mm. While avoiding the main veins of the leaf, three points, the upper, middle, and lower, were uniformly selected along the direction of the main veins of the leaf, and LT was the average of the three points. Leaves after LFW and LT measurements were put into paper envelopes, placed in an oven at 75 °C, dried for 72 h to a constant weight, weighed, recorded leaf dry weight (LDW). specific leaf area (SLA), leaf dry matter content (LDMC), and leaf tissue density (LTD) trait metrics were calculated according to the following equations: SLA = LA/LDW; LDMC = LDW/LFW; LTD = 1/(SLA×LT). The dried leaves were subsequently pulverized, sieved through a 40 mesh sieve and then the carbon, nitrogen and phosphorus contents of the leaves were determined using an elemental analyzer (Euro Vector EA3000).

We chose two diversity indicators, tree species diversity (i.e., richness) and functional diversity, to examine their effects on AGB in each 20 m × 20 m plot. Functional diversity is characterized by functional divergence, which is a measure of the maximum dispersion in trait space of the multidimensional distribution of functional traits in a community (Mason et al. 2005). The trait indices used to calculate functional divergence include SLA, LDMC, LTD, leaf carbon content (LCC), leaf nitrogen content (LNC) and leaf phosphorus content (LPC). We measured these functional traits for a total 119 tree species. Based on the frequency of tree species occurrence in each 20 × 20 m plot, we calculated community functional diversity and the community-weighted mean (CWM) for each trait using the R package 'FD' (Laliberté and Legendre 2010).

Stand structural attributes

For each 20 m × 20 m plot, we calculated five structural variables: stand density (i.e., number of individual trees per ha), maximum DBH, coefficient of DBH variation (DBH_cv), height coefficient of variation (Height_cv), and canopy cover, where DBH_cv is the ratio of the standard deviation to the mean of DBH for all trees in the community. Height_cv and canopy cover were calculated after scanning by the D2000S quadrotor unmanned aerial vehicle LiDAR (Shenzhen Feima Robotics Technology Limited Liability Company). The height variable is a statistical parameter calculated using the point cloud elevation values. The point cloud space was divided into 20 m × 20 m rasters. The height variance coefficient within each 20 m × 20 m raster was calculated as: Height_cv = $(Z_{std}/Z_{mean}) \times 100\%$, where Z_{std} is the standard deviation of the height values of all points obtained and Z_{mean} is the average height of all points obtained. Canopy cover is the vertical projection of the stand canopy as a percentage of the forest floor area, and is an important factor in responding to forest structure and environment. Coverage estimation is based on the small spot system by calculating the ratio between the number of echoes from vegetation and the number of echoes from the ground. The point cloud space was divided into different rasters according to the set resolution (20 m × 20 m), and the canopy cover of each raster was obtained by calculating the ratio between the number of vegetation points and the total number of points. Points larger than the height threshold (2 m) were considered as vegetation points in the calculation process with the formula: $CC = n_{veg}/n_{total}$, where CC is canopy cover, n_{veg} is the number of vegetation points, and n_{total} is the total number of points. The LiDAR data were acquired during the defoliation period, October 2023, a period that minimizes errors in tree cover below the canopy due to leaf shading.

Statistical analysis

Firstly, AGB data was ln-transformed for all analyses. Spearman correlation coefficients were used to explore bivariate relationships between biomass and four types of variables: (1) environmental factors, including topographical factors and soil factors; (2) diversity indicators, including tree species richness, functional diversity, (3) the community-weighted mean of each trait index; and (4) stand structural variables, including horizontal structure and vertical structure. The Spearman rank correlation coefficient does not require the assumptions of normality and homoscedasticity for the variables, and it is less sensitive to outliers. Therefore, its applicability is generally broader.

We then used ordinary least squares (OLS) multiple regression models to examine the effects of the three

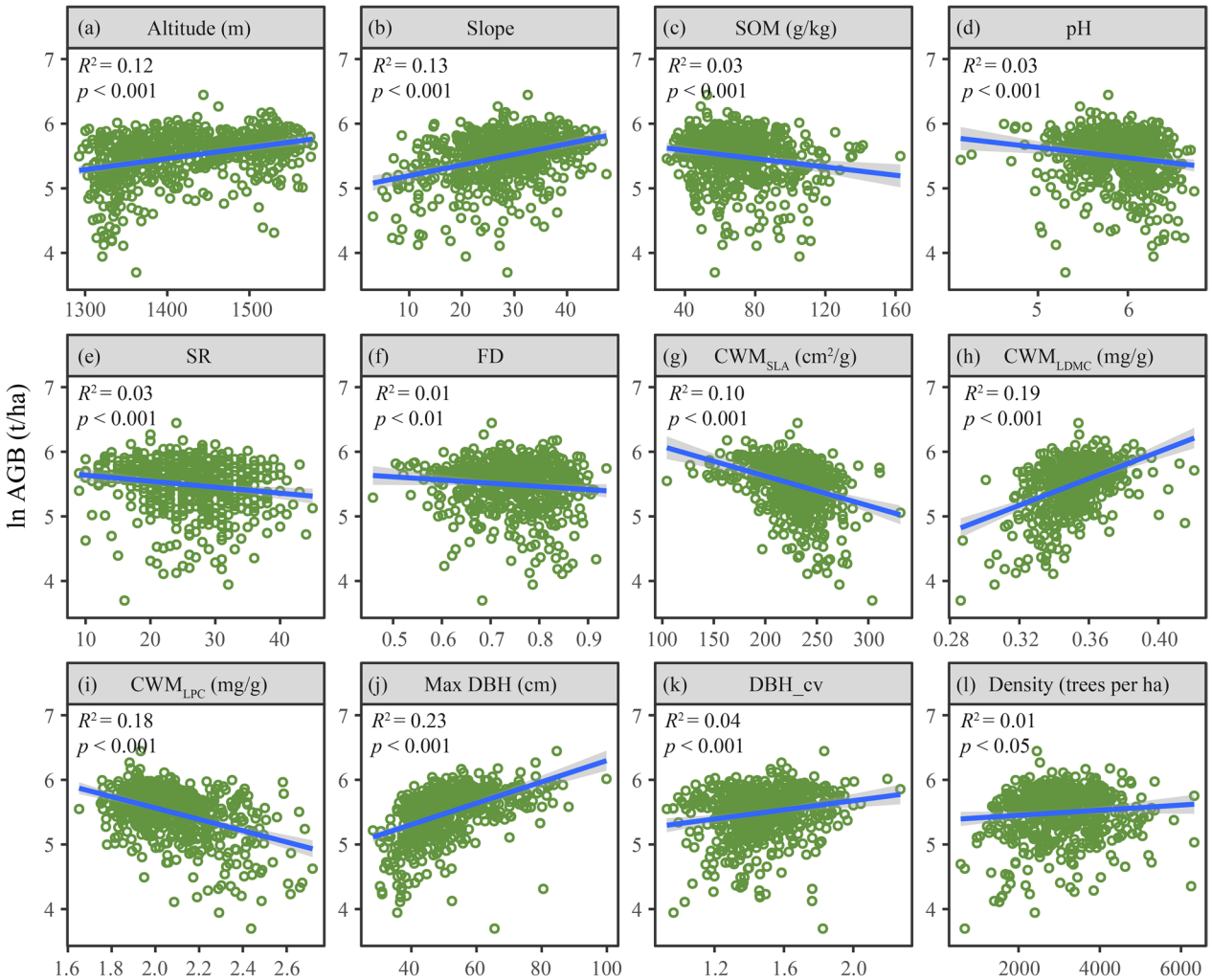


Fig. 1 Bivariate relationships between above-ground biomass (AGB, ln-transformed) and topography [(**a**) altitude and (**b**) slope], soil [(**c**) soil organic matter (SOM) and (**d**) soil pH], biodiversity [(**e**) species richness (SR), (**f**) functional diversity (FD)], community-weighted mean trait value of [(**g**) specific leaf area (CWM_{SLA}), (**h**) leaf dry matter content (CWM_{LDMC}) and (**i**) leaf phosphorus content (CWM_{LPC})] and stand structure [(**j**) maximum diameter at breast height (max DBH), (**k**) DBH variation (DBH_cv) and (**l**) stand density] in the warm temperate deciduous broad-leaved forest ($N = 625$, these points refer to the values of the different indicators in each individual 20 m × 20 m plot)

types of variables on biomass. To remove the multicollinear variables, we included variables with the variance inflation factor (VIF) < 3 in the analysis of the multiple regression model. The optimal model was chosen using all-subsets regression analysis. To assess the relative significance of each variable, we performed the random forest model analysis using package of "randomForest" in R (Liaw and Wiener 2002). The random forest model is capable of effectively managing datasets with high-dimensional features without requiring dimensionality reduction, and it can also assess the relative importance of variables. In this model, the ranking method is based on the percentage of increase of mean square error (increase in MSE (%)).

Finally, we used structural equation model (SEM) to examine the direct or indirect effects of variables on AGB. We evaluated the overall fit of the SEM using goodness-of-fit index (GFI), standardized root mean square residual (SRMR) and p-value (where GFI > 0.95, SRMR < 0.05, $p > 0.05$ indicate a good fit). The SEM model was performed using the R package "lavaan" (Rosseel 2012). All analyses were performed using the R software version 4.2.3 (R Core Team 2023).

Table 1 Summary of the best ordinary least squares (OLS) multiple regression model for the effects of different types of variables on AGB ($N = 625$)

Variable	Estimate	*SE*	*t*-value	*p*-value	VIF
Slope	0.237	0.033	7.22	< 0.001	1.117
Convexity	0.194	0.031	6.22	< 0.001	1.009
FD	−0.089	0.033	−2.72	< 0.010	1.118
CWM_{LNC}	−0.152	0.033	−4.62	< 0.001	1.129
Density	0.153	0.032	4.77	< 0.001	1.073
Max DBH	0.405	0.033	12.40	< 0.001	1.110
Height_cv	−0.116	0.032	−3.68	< 0.001	1.035
Canopy cover	0.092	0.032	2.85	< 0.010	1.092
	DF	SE_{resid}	*F*-value	*p*-value	R^2
Model statistics	616	0.280	52.880	< 0.001	0.407

SE standard error, *VIF* variance inflation factor, *FD* functional diversity, CWM_{LNC} community-weighted mean trait value of leaf nitrogen content, *Max DBH* maximum diameter at breast height, *Height_cv* height variation, *DF* degree of freedom, SE_{resid} residual standard error

Results

Relationships between AGB and individual predictor variables

In the 25-ha plot of Qinling Huangguan, AGB was highly variable, ranging from 40.36 to 629.70 t/ha, with a mean value of 257.65 t/ha. For the topographic and soil variables, AGB had a significant positive correlation with three topographic variables (altitude, slope and convexity), however, a negative relationship with soil factors. Specifically, slope ($R^2 = 0.13$, $p < 0.001$, Fig. 1) had the strongest positive relationship with AGB, followed by altitude ($R^2 = 0.12$, $p < 0.001$, Fig. 1), convexity ($R^2 = 0.05$, $p < 0.001$, Fig. S1), but aspect was not significantly correlated ($R^2 < 0.03$, $p < 0.001$, Fig. S1). AGB showed weak negative correlations with SOM, pH and TN, but no significant correlation with TP ($R^2 < 0.001$, $p > 0.05$, Fig. S1).

When investigating the bivariate relationships between AGB and diversity, we found a significant negative correlation between tree species richness and AGB ($R^2 = 0.03$, $p < 0.001$, Fig. 1). Among functional traits, CWM_{SLA}, CWM_{LNC} and CWM_{LPC} were significantly and negatively correlated with AGB ($R^2 > 0.05$, $p < 0.001$, Fig. S2). Nevertheless, CWM_{LDMC}, CWM_{TD} and CWM_{LCC} were significantly and positively correlated with AGB ($R^2 > 0.02$, $p < 0.001$, Fig. S2).

Most of the stand structural variables had positive individual correlation with AGB, especially, the strongest positive correlation with AGB was maximum DBH ($R^2 = 0.23$, $p < 0.001$, Fig. 1), followed by coefficient of DBH variation ($R^2 = 0.04$, $p < 0.001$, Fig. 1) and canopy cover ($R^2 = 0.02$, $p < 0.001$, Fig. S3). The relationship between AGB and stand density was relatively weak ($R^2 = 0.01$, $p < 0.05$, Fig. 1). There was a weak negative correlation between height_cv and AGB ($R^2 < 0.01$, $p < 0.05$, Fig. S3).

Effects of biological and abiotic factors on AGB

In the multiple regression model, we first pick out the variables with VIF < 3 to remove multicollinearity and then determined the optimal model based on all-subsets regression analysis. In the best model, a total of eight variables were selected, including slope, convexity, FD, CWM_{LNC}, stand density, maximum DBH, height coefficient of variation and canopy cover ($R^2 = 0.41$, $SE_{r\text{-}esid} = 0.28$, $p < 0.05$, Table 1).

According to random forest model, the predictive model (explaining 55% of the variation) suggested that the above factors are importance in explaining the variance in driving AGB. Of these, maximum DBH (38.66%) and CWM_{TD} (23.65%) explained the largest variation in AGB, followed by DBH_cv, CWM_{LPC} and CWM_{LDMC}, and then by altitude, stand density and slope (Fig. 2).

Direct and indirect effects of drives on AGB

Soil chemical indicators were particularly low in explaining variation in AGB, both in terms of correlation and significance, so soil variables were excluded from the determination of direct or indirect effects. Ultimately, we incorporated slope, functional diversity, maximum DBH and stand density into structural equation model. The SEM was well fitted to the data for these variables ($p(\chi^2) = 0.58$, GFI = 0.998, SRMR = 0.011, Fig. 3) and explained 33% of variance in AGB. Specifically, the result showed a direct positive effect of stand structure indicators on AGB. In addition, functional diversity indirectly affected AGB via stand density (Fig. 3). Maximum DBH also indirectly affected AGB via stand density, whereas had a negative effect on stand density (Fig. 3).

Discussion

Selection effects: significant effects of functional traits on AGB

Most studies have found that biodiversity is generally positively correlated with community productivity and AGB (Wagg et al. 2014; Jing et al. 2015; Liang et al. 2016; Chen et al. 2018; Huang et al. 2018). Two hypotheses that could account for this positive relationship are niche complementarity effects and selection effects. In contrast, our study revealed a significant negative bivariate relationship between species richness and AGB. Our study revealed a significantly negative bivariate relationship between species richness and AGB (Fig. 1), which aligns with some research (Jacob et al. 2010), but contrary to more studies (Cavanaugh et al. 2014; Tilman et al. 2014; Chen et al. 2018; Ouyang et al. 2019; Augusto and Boča 2022). The negative correlation observed between

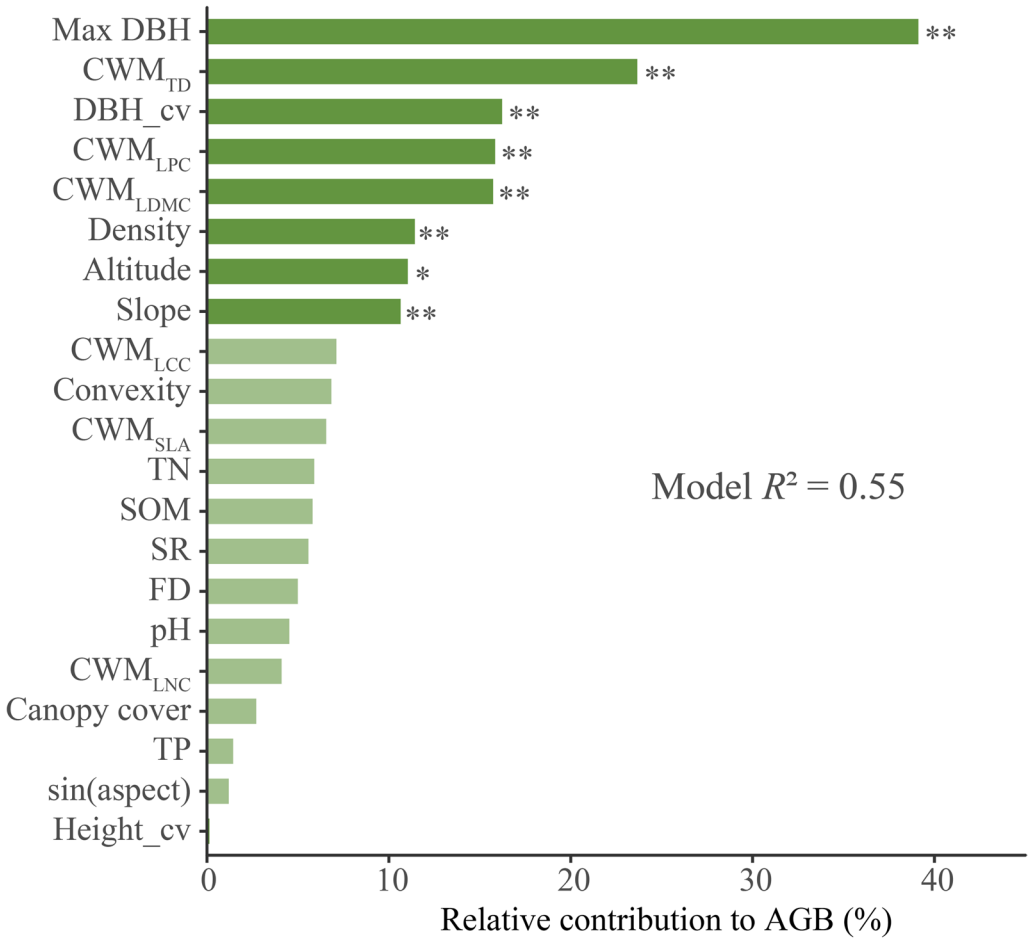


Fig. 2 Relative contribution to above-ground biomass (AGB) explained by topography [altitude slope, convexity and aspect], soil [soil organic matter (SOM), soil pH, soil total nitrogen (TN) and soil total phosphorus (TP)], biodiversity [species richness (SR) and functional diversity (FD)], community-weighted mean trait value of [specific leaf area (CWM_{SLA}), leaf dry matter content (CWM_{LDMC}), leaf carbon, nitrogen and phosphorus content (CWM_{LCC}, CWM_{LNC}, CWM_{LPC})] and stand structure [maximum diameter at breast height (max DBH), DBH variation (DBH_cv), stand density, canopy cover and height variation (Height_cv)] in the random forest model. The graph shows the significance scores for each predictor variable, where significant scores (The significance levels are: ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$ and no symbol $p \geq 0.05$) are shown in dark green and non-significant scores are shown in light green. The R^2 denotes the variance explained by the full model

AGB and species richness suggests that the niche complementarity effect was not evident in our study. Besides, species richness became non-significant when other variables were accounted for in both the multiple regression model (Table 1) and random forest model (Fig. 2). Additionally, we did not find evidence that functional diversity enhances AGB (Figs. 2, 3), further supporting the notion that niche complementarity effects may not be significant in this study area.

Some studies on biodiversity-ecosystem functioning (BEF) and biodiversity-ecosystem multifunctionality (BEMF) primarily utilize species richness as a metric of biodiversity (Wu et al. 2015; Huang et al. 2018; Chen et al. 2019). Nevertheless, with a deeper investigation into diversity, an increasing array of diversity metrics, such as variations in functional traits, has been shown to serve as more accurate predictors of ecosystem functioning compared to species richness (Gross et al. 2017; Vargas-Larreta et al. 2021; Li et al. 2022). In our study, the random forest result showed that the CWM of LTD, LPC, and LDMC contributed more to the variation of AGB than species richness (Fig. 2), which verifies that selection effects play an important role in increasing forest biomass. Specifically, the CWM of LTD and LDMC exert a positive influence on AGB. Plants exhibiting high LDMC and large LTD are typically more robust and resistant to physical stress (Cornelissen et al. 2003). Meanwhile, LTD reflects the accumulation of biomass in plant organs (Reich et al. 1998), and in this study, LDMC was positively correlated with LTD (Fig. S4), implying

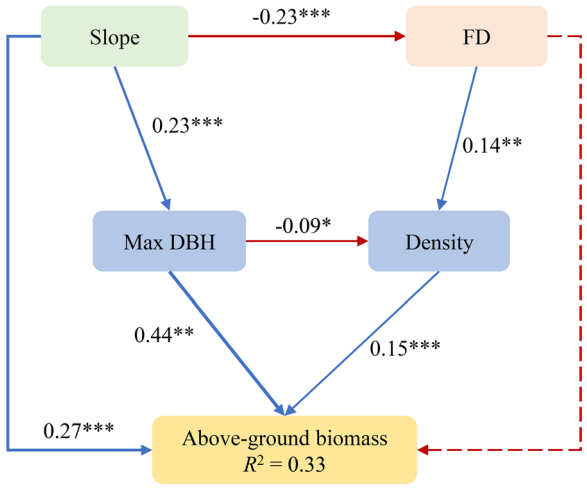


Fig. 3 Structural equation model on above-ground biomass (AGB) to slope, functional diversity (FD), maximum diameter at breast height (max DBH) and stand density in the warm temperate deciduous broad-leaved forest. Model-fit statistics: $\chi^2 = 1.95$, $df = 3$, $p = 0.58$, GFI = 0.998, RMSEA = 0.011. The blue lines indicate positive effects, the red lines indicate negative effects; the solid lines indicated significant and the dashed line indicates non-significant

that more leaf photosynthesis products were used to accumulate dry matter. Besides, high LDMC endows plants with structural integrity and enhances their survival in challenging environments, such as low temperatures (Oktavia and Jin 2020). Increased LDMC leads to higher leaf biomass, facilitating water retention in leaves and improving the plant's ability to withstand intense light (Donovan et al. 2011), which is consistent with a conservative resource utilization strategy (Matthews et al. 2017). On the other hand, it has been shown that LPC is correlated with plant size, with larger individuals exhibiting lower leaf phosphorus content (Gao et al. 2019; Qiu et al. 2022), which can also explain the negative correlation between LPC and AGB.

The relationships between environmental variables and AGB

Topographic factors play a crucial role in redistributing elements essential for plant growth and indirectly influence AGB accumulation within communities (Johnson et al. 1996; Takyu et al. 2003). For instance, AGB is influenced through the regulation of water and nutrient availability and uptake by plants (Jucker et al. 2018), as well as by shaping the microenvironment in which trees develop, thereby affecting species composition (Scholten et al. 2017). In this study, AGB was primarily influenced by altitude and slope. AGB was primarily influenced by altitude and slope. The positive effect of altitude on AGB primarily reflects how altitude impacts the distribution of tree sizes, with a trend toward larger individuals at higher altitudes. Terrain concavity, meanwhile, is a key factor influencing precipitation retention time; greater terrain concavity enhances water retention, prolonging water availability for trees and thereby facilitating biomass accumulation (Engelbrecht et al. 2005). In addition, we found that soil nutrients had non-significant effects on AGB, potentially due to nutrient imbalances in certain species or their adaptive responses to the surrounding environment (Prado-Junior et al. 2016; Poorter et al. 2017; Ali et al. 2019).

Stand structural attributes are better predictors of AGB than species richness

In this study, tree species richness showed no significant correlation with AGB, likely due to asymmetrical competition based on individual size and the competition for limited resources, which may diminish the value of niche complementarity. The impact of diversity on ecosystem functioning may be primarily shaped by factors such as the selection effects of dominant species and environmental conditions. Here, the maximum tree and stand density had a greater effect on AGB than tree species richness (Fig. 2). This result suggests that forest AGB is driven by the characteristics of dominant tree species. The disproportionate contribution of large trees to biomass relative to other single variables supports the hypothesis of the "big-sized tree hypothesis". The scaling theory indicates that large trees, which are often taller, possess larger crowns and occupy the top of the canopy, capture more light resources to outcompete neighboring trees (West et al. 2009; Van Pelt et al. 2016; Xu et al. 2016; Ligot et al. 2018). In addition, our findings revealed a positive relationship between stand density and biomass (Fig. 3). The random forest and SEM analyses further indicted that stand density directly increased AGB (Fig. 2, 3). It is now thought that species richness indirectly affects forest productivity by influencing stand density (Ouyang et al. 2019). Typically, higher stand density intensifies tree interactions, and a diverse forest structure promotes complementarity (Boyden et al. 2005), as trees occupy more space and utilize more resources (Forrester et al. 2019), thus enhancing forest productivity. However, there may also be a negative correlation between stand structural attributes and forest functions due to competitive exclusion. Recent research has shown a significant negative correlation between structural diversity and biomass increment (Zhai et al. 2024). With increasing

structural diversity, tree mortality rises (Zhai et al. 2022) and the growth of smaller trees is inhibited (Looney and Zhang 2022), resulting in a negative effect on net biomass change. In summary, variations in stand structure drive individual interactions and can positively or negatively influence forest ecosystem functioning.

Conclusions

Most studies related to the prediction and assessment of ecosystem functioning have focused on the effect of biodiversity on it. Our study showed that stand structure (including maximum DBH and stand density), community-weighted trait means, and environmental factors (i.e., elevation and slope) all contributed to geographic variation in AGB, with each exerting a greater influence on AGB than species richness. These findings indicate that stand structure attributes may be more effective predictors of forest AGB compared to species richness. Compared to species richness and other biodiversity indicators, the role of stand structure in forest ecosystem functioning remains understudied. Therefore, future research should enhance our understanding of the relationship between stand structure and plant dynamics, particularly exploring how stand structure affects individual recruitment, growth, and mortality, as well as interactions among individuals.

Abbreviations

AGB	Above-ground biomass
Max DBH	Maximum diameter at breast height
DBH_cv	Coefficient of variation of diameter at breast height
Height_cv	Coefficient of variation of height
SOM	Soil organic matter
TN	Soil total nitrogen content
TP	Soil total phosphorus content
LFW	Leaf fresh weight
LT	Leaf thickness
LA	Leaf area
LDW	Leaf dry weight
SLA	Specific leaf area
LDMC	Leaf dry matter content
LTD	Leaf tissue density
LCC	Leaf carbon content
LNC	Leaf nitrogen content
LPC	Leaf phosphorus content
SR	Species richness
FD	Functional diversity
CWM	Community-weighted mean

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13717-024-00569-7.

Additional file 1.

Additional file 2.

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Author contributions

Qiulong Yin, Zhanqing Hao and Chunmei He designed the research. Zhanqing Hao provided project support. Chunmei He, Yushan Li and Xiaoxia Dai organized the dataset and analyzed the data. Chunmei He wrote the manuscript with the assistance of Zhanqing Hao, Qiulong Yin and Zuoqiang Yuan. Na Liu, Fangfang Wu, Yonghui Liang and Meiping Gao conducted the fieldwork and collected the data. Jiangbo Yan provided important information on AGB calculations. All authors gave final approval for publication.

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Availability of data and material

The data sets used in the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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